

Hw7 con.

$$\begin{array}{l} 2. \quad x+y=5 \\ \quad \quad 2x-y=16 \end{array} \rightarrow \text{Added eq.}: 3x=21$$

$$x=7$$

$$y=5-7=-2$$

$$\text{Solution: } (x,y) = (7,-2)$$

$$3. \quad -x+y+z=2$$

$$x-y-z=1$$

$$x+y-z=0$$

Adding first 2 equations:

$$(-x+y+z) + (x-y-z) = 2+1 \Rightarrow 0=3$$

$0=3 \rightarrow$ False/contradiction as $0 \neq 3$

No Solution

$$4. \quad 2x+2y+2z=4$$

$$2x-y+2z=4$$

$$-x+2y-z=-2$$

$$\rightarrow 2x-y+2(2-x-y)=4$$

$$\hookrightarrow 2x-y+4-2x-2y=4$$

$$\hookrightarrow -3y+4=4 \Rightarrow -3y=0 \Rightarrow y=0$$

$$2x+2y+2z=4 \rightarrow x+y+z=2 \rightarrow z=2-x-y$$

$$\rightarrow -x+2(0)-(2-x-0)=-2$$

$$\hookrightarrow -x-2+x=-2$$

$$-2=-2 \checkmark \text{ Works for all } x \text{ (consistent)}$$

$y=0, z=2-x$: Infinitely many solutions

$$(x,y,z) = (2-z, 0, z)$$

$$\begin{array}{l} 5. \quad \text{Ball} = b \\ \quad \text{Bat} = b + 1.00 \\ \quad \text{Total} = 1.10 \end{array}$$

$$b + (b + 1.00) = 1.10 \Rightarrow 2b = 0.10 = b = 0.05$$

$$\text{Bat} = 1.05, \text{Ball} = 0.05$$